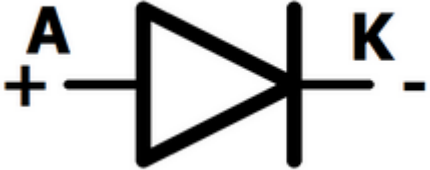
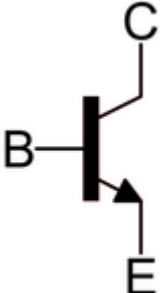
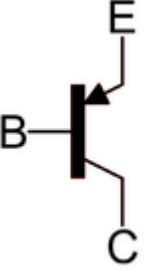
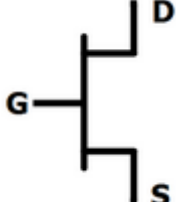
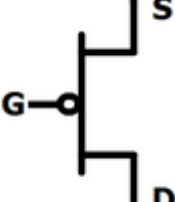


LPRS 1

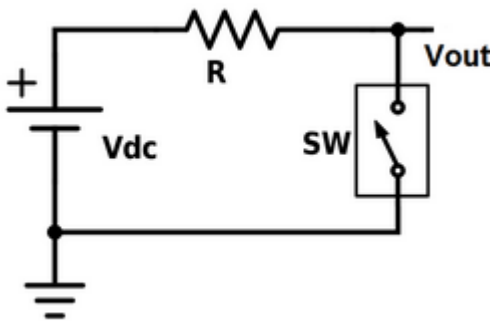
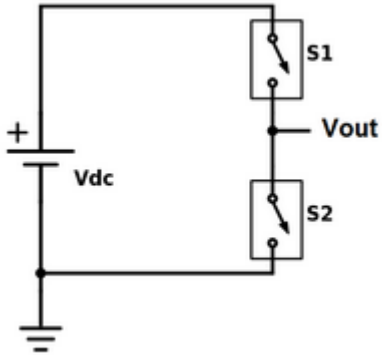
ISPITNA PITANJA - 2019

1. Osnovni prekidački elementi

Dioda	NPN tranzistor	PNP tranzistor
		
$V_{th} \approx 0.6V, V_{ak} \geq V_{th} - \text{dioda je uključena}$	$V_{be} \geq V_{th}$	$V_{be} \leq - V_{th} $

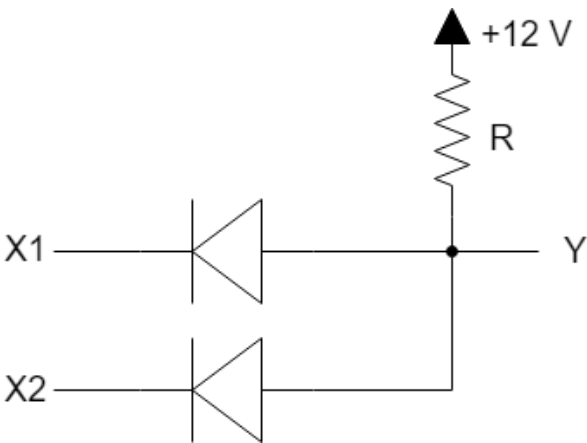
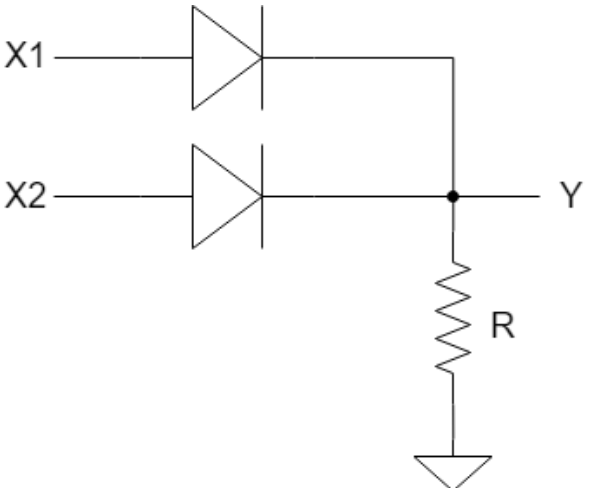
N kanalni MOSFET	P kanalni MOSFET
	
$V_{gs} \geq V_{thMOS}$	$V_{gs} \leq - V_{thMOS} $

2. Osnovna prekidačka kola

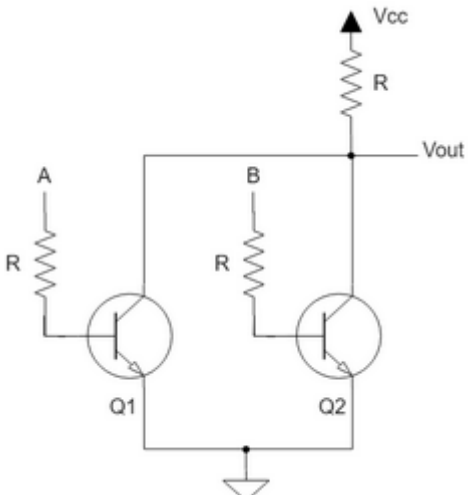
Osnovno prekidačko kolo	CMOS osnovni koncept
	
$SW \text{ otvoren} \implies V_{out} = V_{dc}$	$S_1 = \text{not}(S_2)$
$SW \text{ zatvoren} \implies V_{out} = 0$	$I = 0A \implies P = 0W \text{ u statičkom režimu}$

3. Realna logička kola

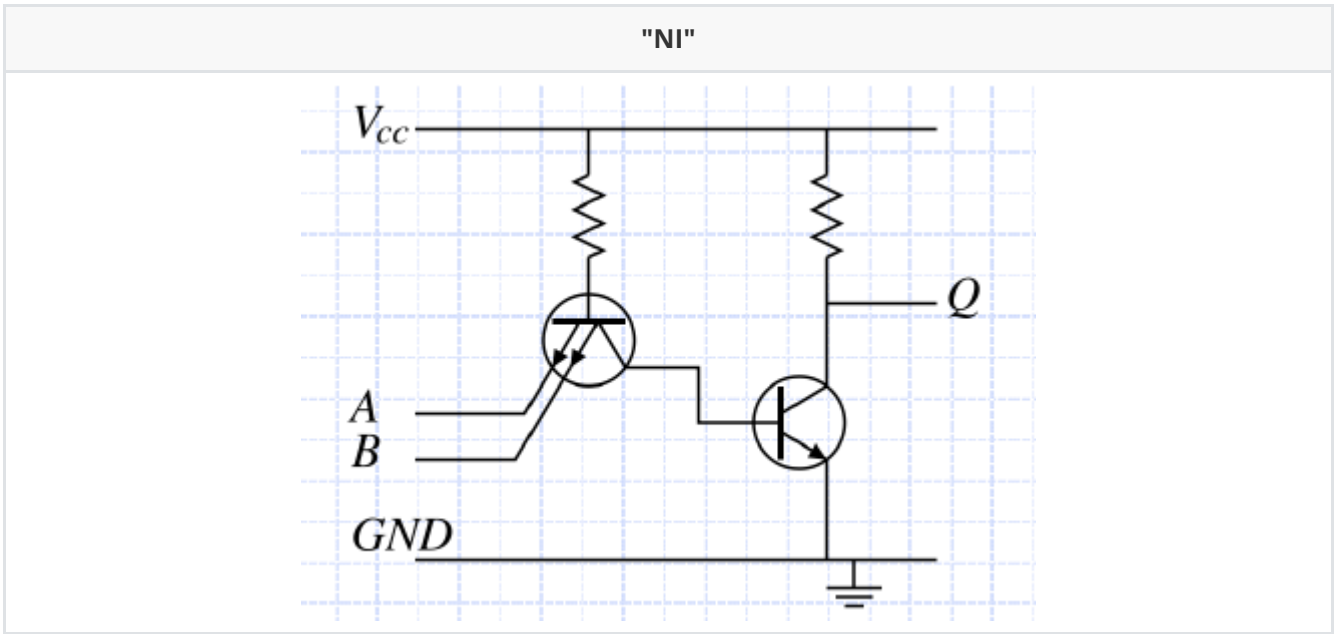
DRL (Diodno-otpornička logička kola)

"I"	"ILI"
	
$Y = X_1 \cdot X_2$	$Y = X_1 + X_2$

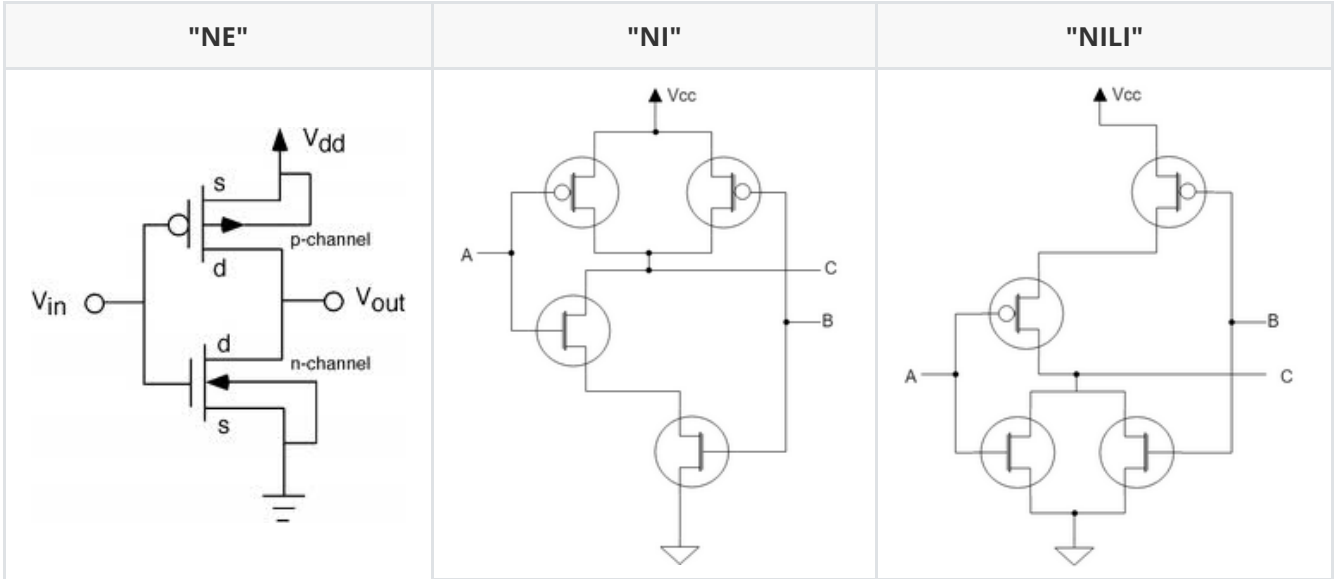
RTL (Resistor-Transistor Logic)

"NILI"


TTL (Transistor-Transistor Logic)



CMOS (Complementary Metal-Oxid-Semiconductor Logic)(NE, NI, NILI)



4. Bulova algebra

[...]

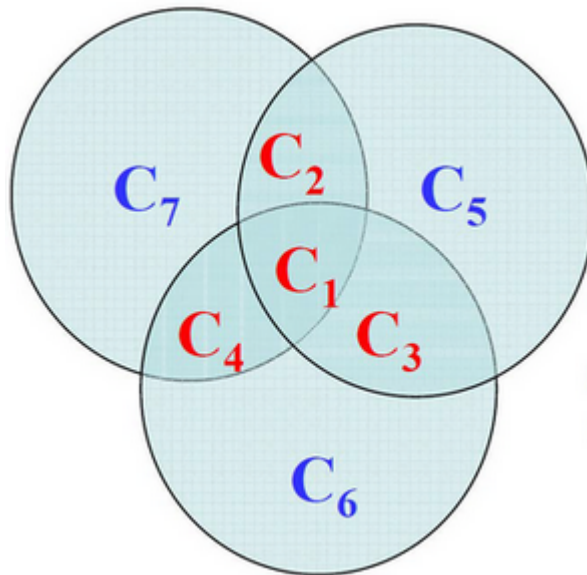
5. Gray-ov kod

Razlika između susednih kodnih reči je 1 bit.

i. sloga	Binarni kod				Grejov kod			
	A	B	C	D	Y3	Y2	Y1	Y0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	1	0
5	0	1	0	1	0	1	1	1
6	0	1	1	0	0	1	0	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	1	0	0
9	1	0	0	1	1	1	0	1
10	1	0	1	0	1	1	1	1
11	1	0	1	1	1	1	1	0
12	1	1	0	0	1	0	1	0
13	1	1	0	1	1	0	1	1
14	1	1	1	0	1	0	0	1
15	1	1	1	1	1	0	0	0

6. Hemingov kod

Predaja



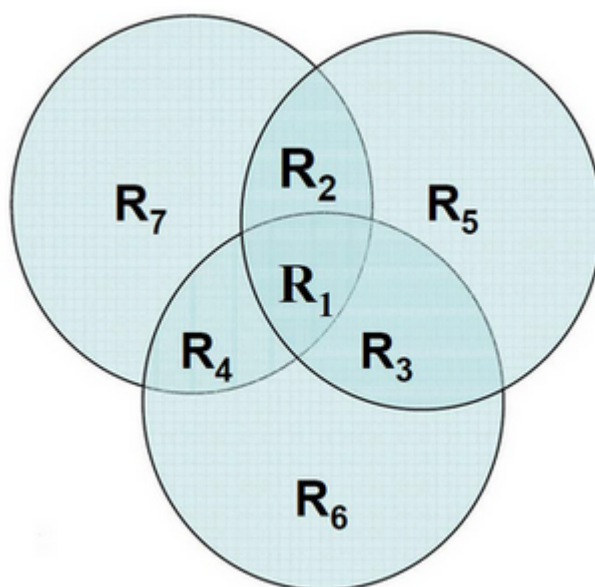
informacija (4 bita) `c1c2c3c4`

dodavanje bita parnosti:

```
c5 <= c1 xor c2 xor c3  
c6 <= c1 xor c3 xor c4  
c7 <= c1 xor c2 xor c4
```

kodna reč: `c1c2c3c4c5c6c7`

Prijem



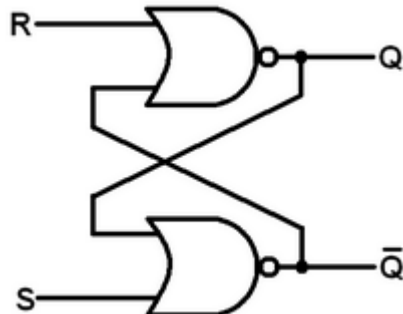
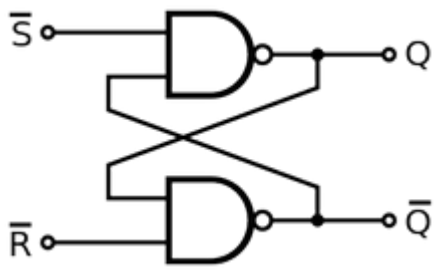
kodna reč: `R1R2R3R4R5R6R7`

detekcija "sindroma":

```
S1 <= R1 xor R2 xor R3 xor R5  
S2 <= R1 xor R3 xor R4 xor R6  
S3 <= R1 xor R2 xor R4 xor R7
```

7. SR Latch (evolucija ka DFF)

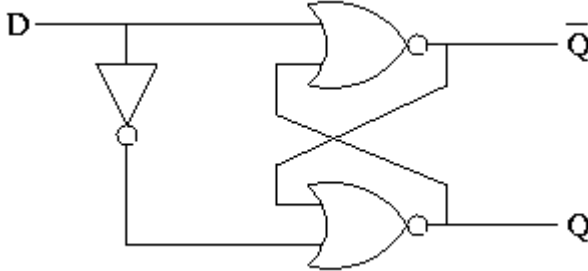
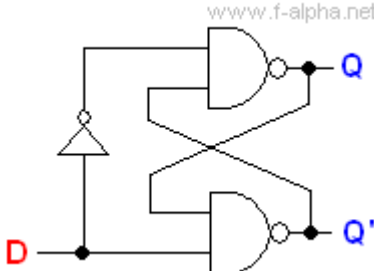
- SR latch

"NILI"	"NI"																																								
																																									
<table><tr><th>S</th><th>R</th><th>Q</th><th>Q̄</th></tr><tr><td>0</td><td>0</td><td>Q-</td><td>Q-</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>	S	R	Q	Q̄	0	0	Q-	Q-	0	1	0	1	1	0	1	0	1	1	0	0	<table><tr><th>S</th><th>R</th><th>Q</th><th>Q̄</th></tr><tr><td>0</td><td>0</td><td>Q-</td><td>Q-</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	S	R	Q	Q̄	0	0	Q-	Q-	0	1	0	1	1	0	1	0	1	1	1	1
S	R	Q	Q̄																																						
0	0	Q-	Q-																																						
0	1	0	1																																						
1	0	1	0																																						
1	1	0	0																																						
S	R	Q	Q̄																																						
0	0	Q-	Q-																																						
0	1	0	1																																						
1	0	1	0																																						
1	1	1	1																																						

Primetimo da su lečevi u zabranjenom stanju kada je ulaz (1,1)

$Q = \bar{Q}$ što je kontradikcija

- D latch

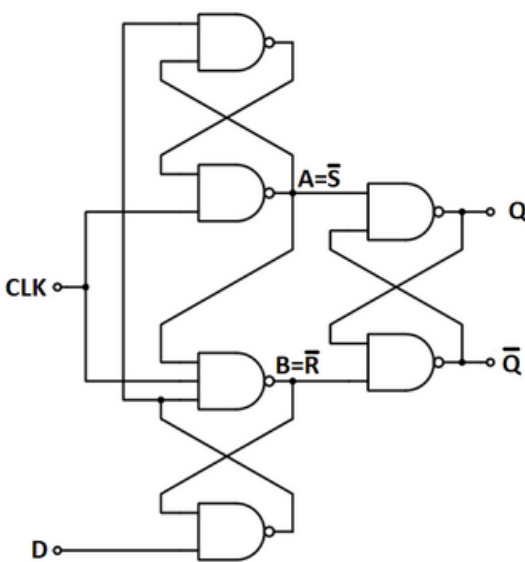
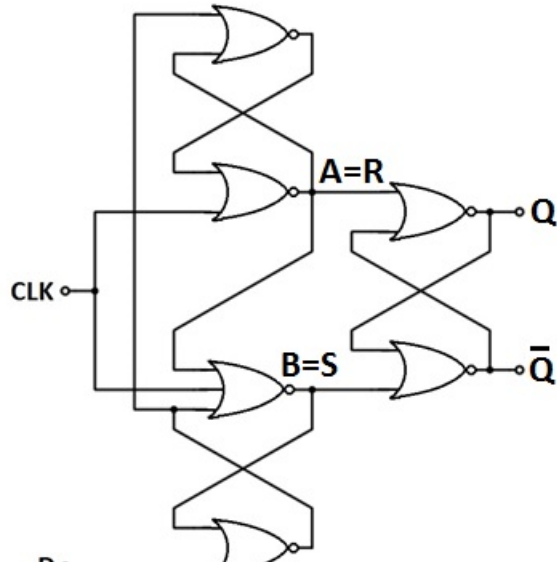
"NILI"	"NI"
	

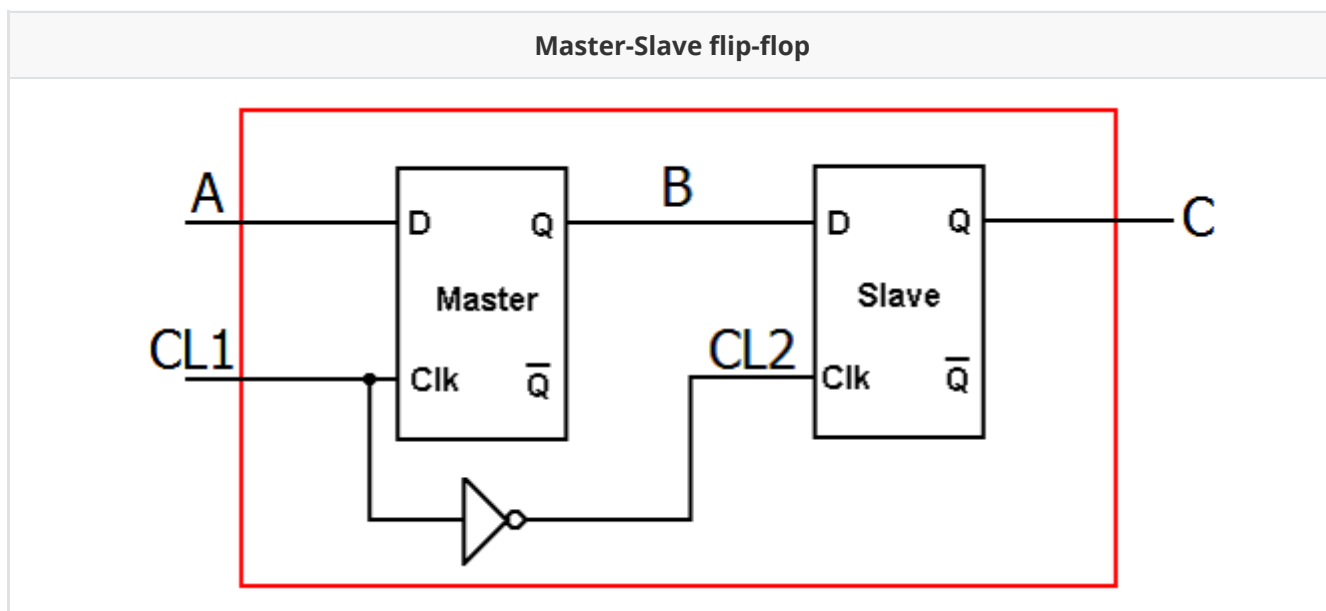
- Gejtovani D latch (Latch sa dozvolom upisa)

The diagram shows a gated D latch circuit. It consists of two NAND gates at the input, followed by two cross-coupled NAND gates forming a SR latch. Input D is connected to the bottom inputs of both input NAND gates. Input E is connected to the top inputs of both input NAND gates through an inverter. The outputs of the input NAND gates are connected to the inputs of the cross-coupled NAND gates. The outputs of the cross-coupled NAND gates are Q and Q-bar.

E	D	Q	$\bar{Q}$
0	0	latch	latch
0	1	latch	latch
1	0	0	1
1	1	1	0

8. Taktovani D flip-flop

sa NI kolima	sa NILI kolima
	
upis na rastuću ivicu clock-a	upis na opadajuću ivicu clock-a



9. Sinhroni/asinhroni reset

Sinhroni reset

```
process (iCLK) begin
    if(iCLK'event and iCLK='1') then
        if(inRST='0') then
            -- postavljanje pocetne vrednosti
        else
            -- upis u registar
        end if;
    end if;
end process;
```

Asinhroni reset

```
process (iCLK, inRST) begin
    if(inRST='0') then
        -- postavljanje pocetne vrednosti
    elsif(iCLK'event and iCLK='1') then
        -- upis u registar
    end if;
end process;
```

10. STD_LOGIC tip

0 - forsirano 0
1 - forsirano 1
X - forsirano nepoznato
L - slabo 0
H - slabo 1
W - slabo nepoznato
Z - velika impedansa
- - don't care
U - nije inicijalizovano

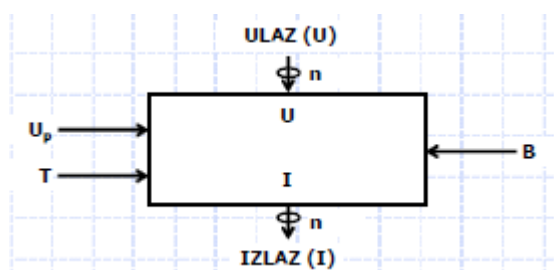
11. Problemi u kombinacionim procesima u VHDL-u

1. Nepokriveni svi slučajevi uslovnih dodela, uzrokuju generisanje lečeva.
2. Kombinaciona petlja, nije dozvoljena jer se gubi vremenski determinizam.
3. Nepotpuna lista osetljivosti dovodi do neslaganja sinteze i simulacije.
4. Multiple driver (isti signal se generiše u više od jednog procesa) – opasnost od kolizije na tom signalu.

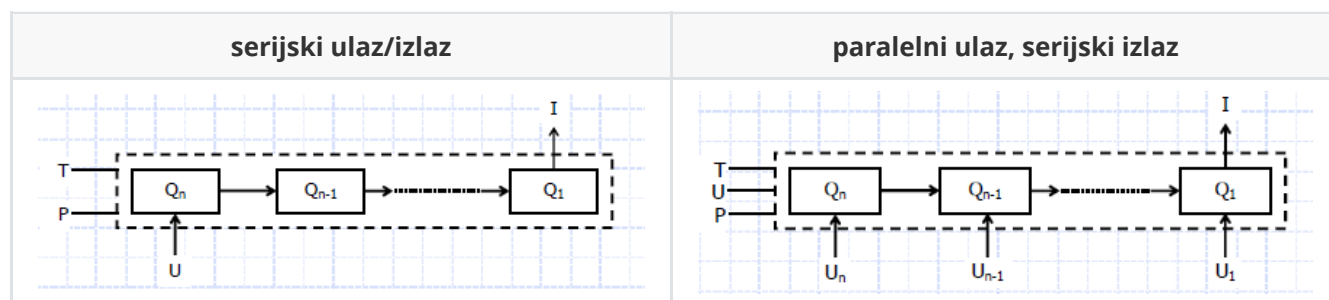
12. VHDL snippeti

...

13. Sekvencijalne mreže (registri, brojači)



- Pomerački registri



- Redni brojač
- Džonsonov brojač

14. Automati

Milijev automat	Murov automat
Milijev automat	
$Z(t) = \lambda(S(t), X(t))$	$Z(t) = \lambda(S(t))$

X – ulazi, Z – izlazi, S – stanje

$Z(t)$ – funkcija izlaza

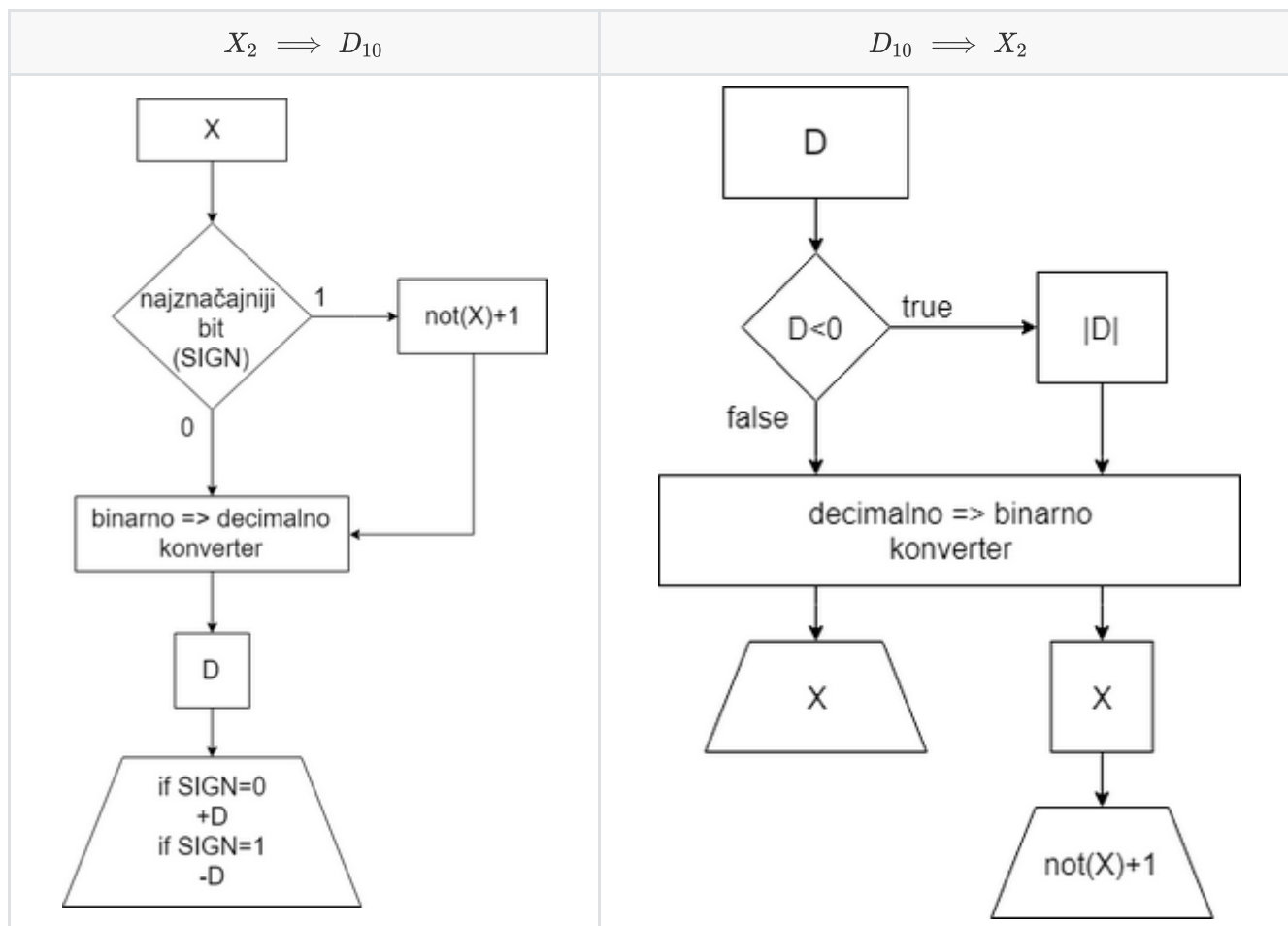
$S(t + \Delta) = \delta(S(t), X(t))$ – funkcija prelaza

[...] trougaona minimizacija stanja automata

15. Predstava brojeva u drugom komplementu

X_2 je binarni broj dat u drugom komplementu

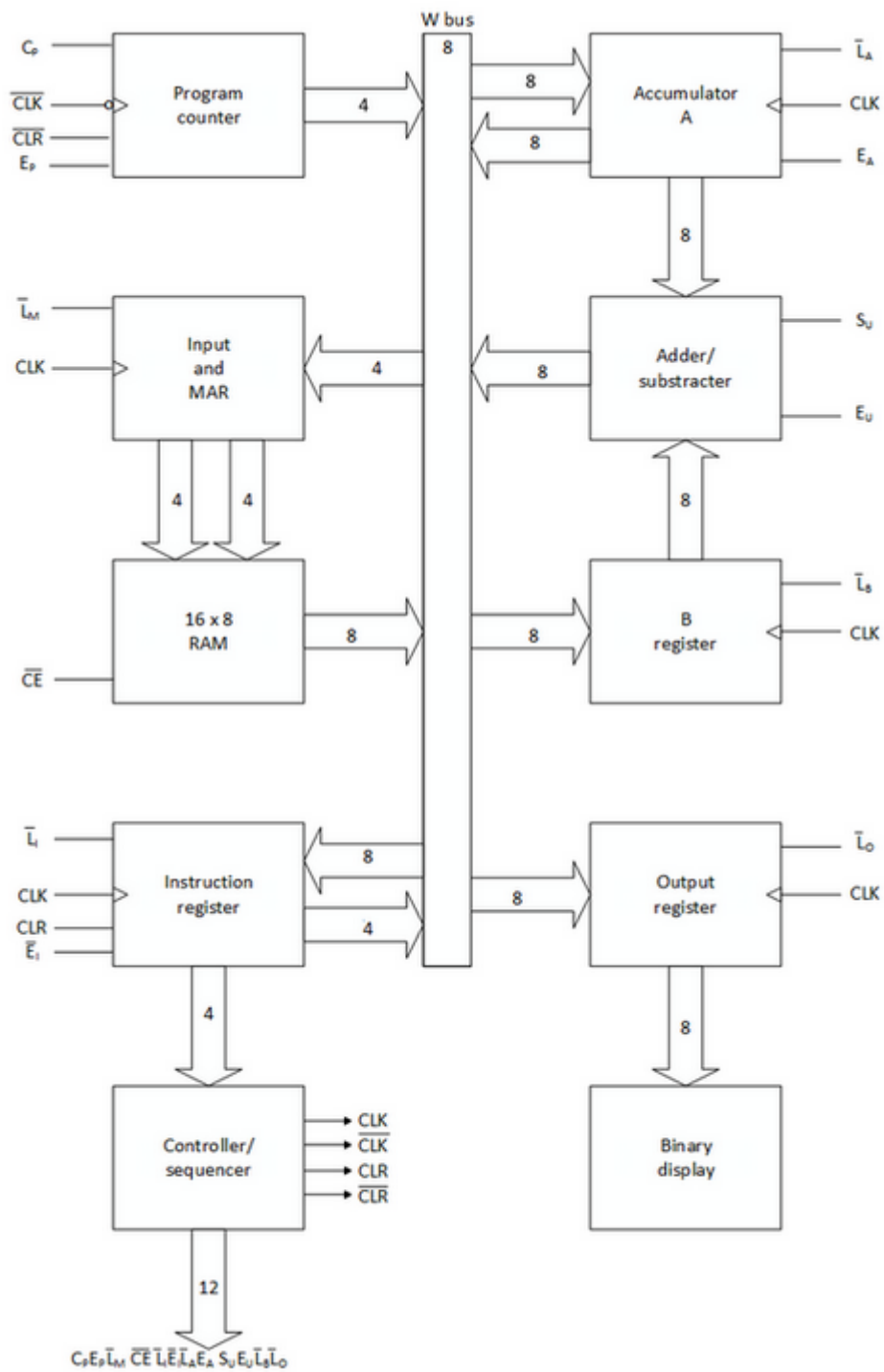
D_{10} je decimalni broj



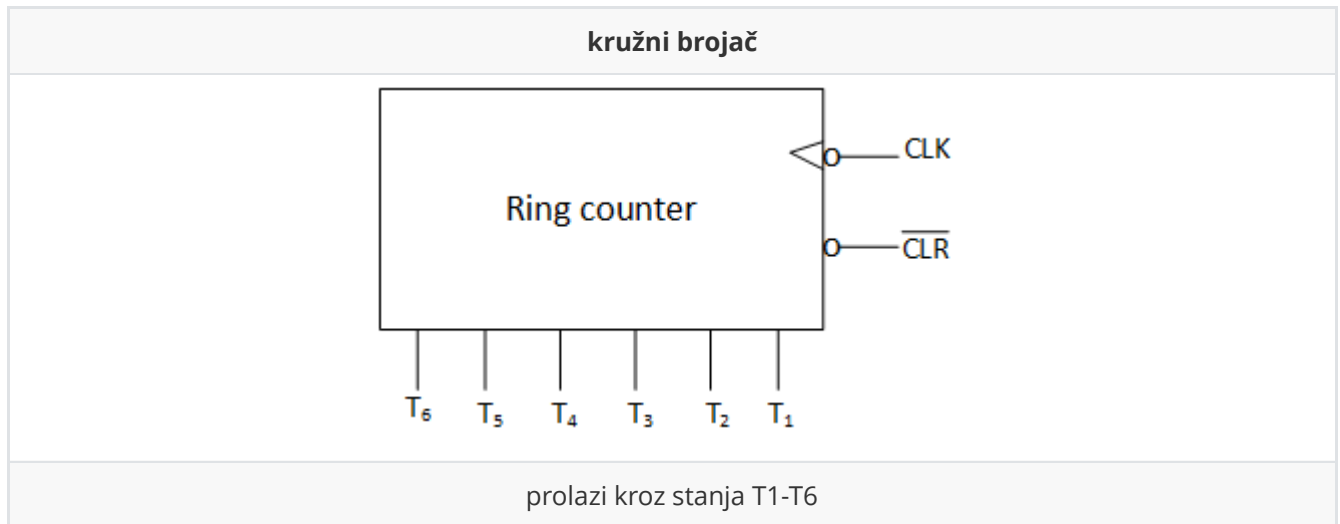
Binarni broj dat u drugom komplementu je proširen za 1 bit (dodavanjem jedinice u izrazu $\text{not}(X) + 1$)

Npr. -3_{10} u drugom komplementu je 101_2 (3 bita), dok 3_{10} je 11_2 .

16. Primer jednostavnog procesora "SAP"



	naredba	kodna reč
LDA	$A \leftarrow RAM$	0000
ADD	$A \leftarrow A + RAM$	0001
SUB	$A \leftarrow A - RAM$	0010
OUT	$OUT \leftarrow A$	1110
HLT	$STOP$	1111



T1, T2, T3 - FETCH (prihvatni ciklus)

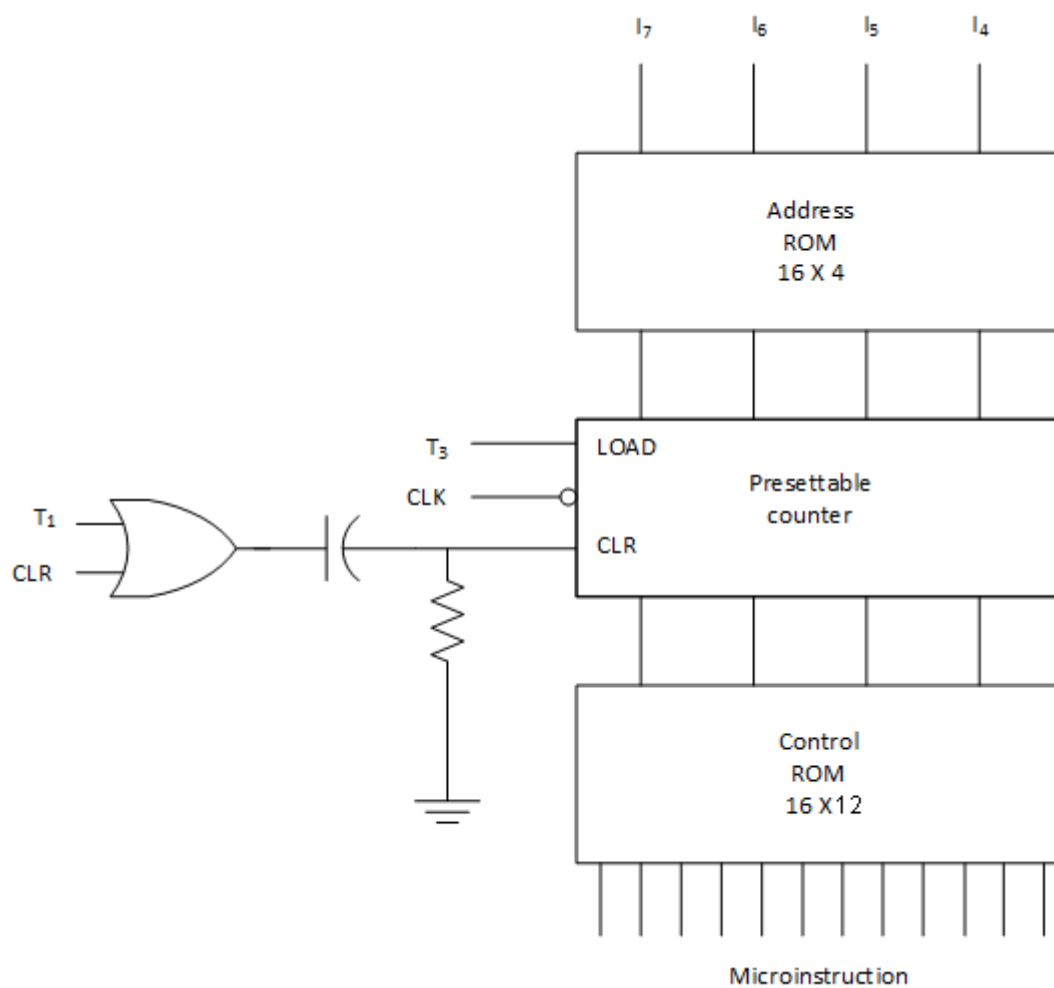
T4, T5, T6 - izvršni ciklus

17. Upravljačka jedinica (tradicionalna hardverska realizacija)

[...]

18. Upravljačka jedinica (mikroprogramska realizacija)

[...]

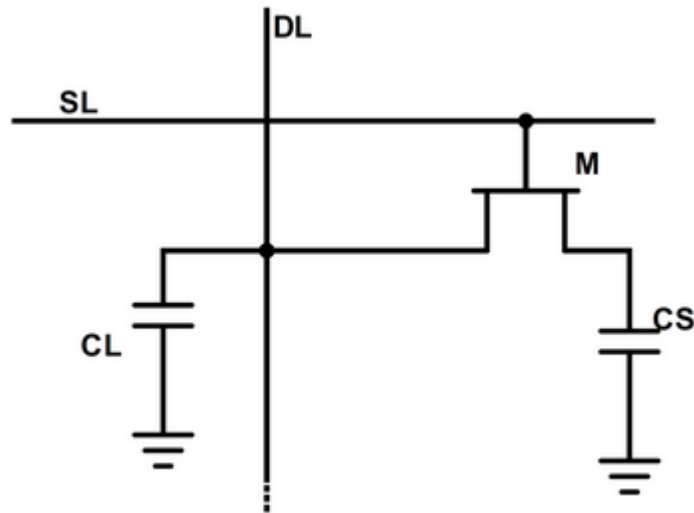


kod naredbe: I₇ I₆ I₅ I₄

19. Logičko projektovanje memorije

a) DRAM (Dynamic Random Access Memory) memorijska ćelija

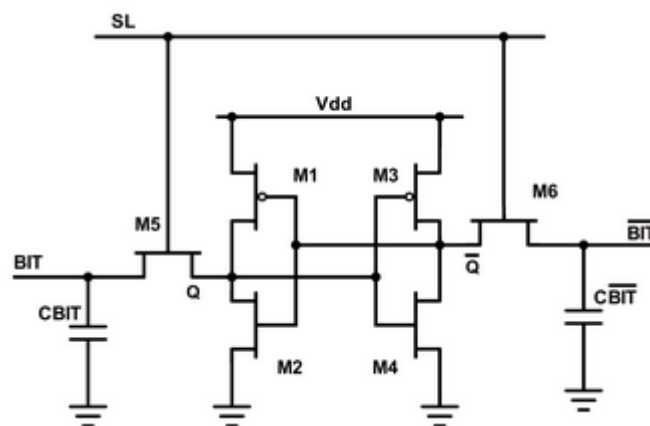
jedan tranzistor po bitu



CL - precharge capacitor
DL - sense amplifier
SL - address selector

b) SRAM (Static Random Access Memory) memorijska ćelija

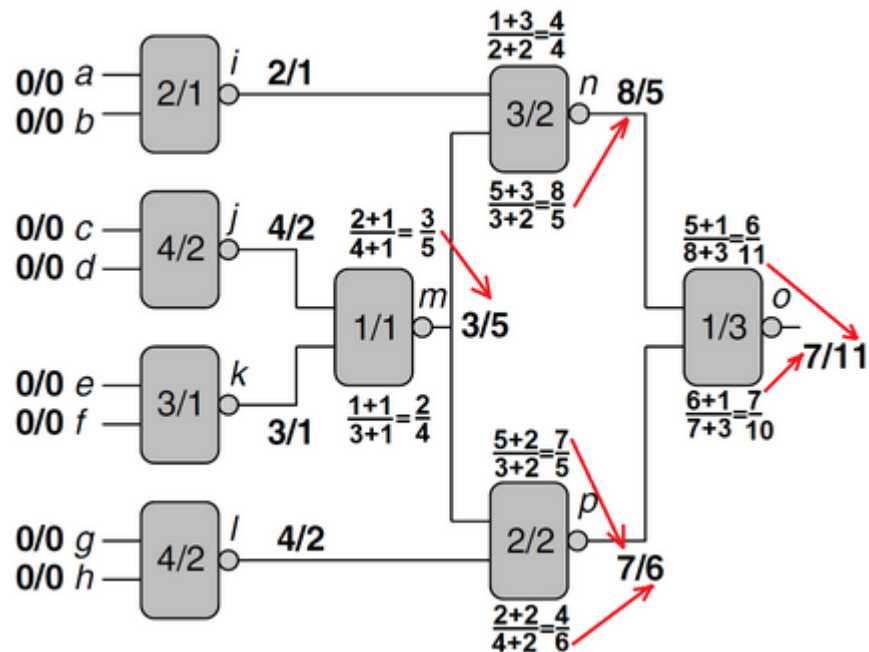
6 tranzistora po bitu približno



20. Vremenske karakteristike digitalnih kola

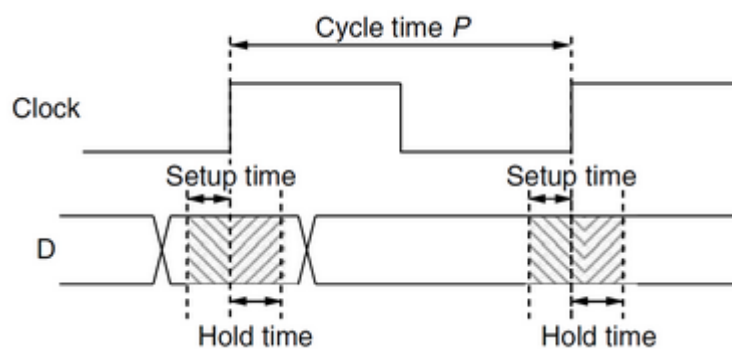
a) Kombinatorna kola - critical path method (CPM)

[rise delay] / [fall delay]



Sušтина je da kada uzmemo razlomak sa ulaza, obrnemo brojeve i dodamo razlomak trenutne logičke kapije (pojedinačno brojilac i imenilac) i na izlaz pišemo najveći broj u brojiocu / najveći broj u imeniocu (ako ima više ulaza).

b) Sekvencijalna kola - setup/hold time constraint



dva flip-flopa: FF_i i FF_j

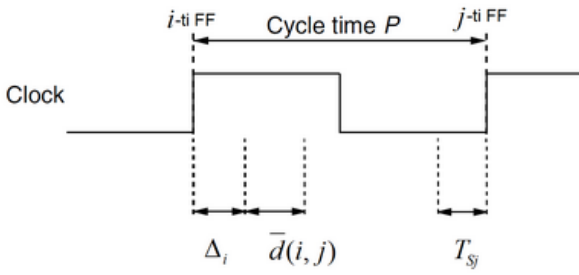
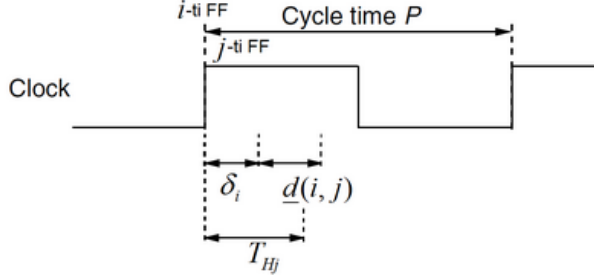
najveće kombinaciono kašnjenje između njih: $\bar{d}_{ij}$

najmanje kombinaciono kašnjenje: $\underline{d}_{ij}$

$$\underline{d}_{ij} \leq d_{ij} \leq \bar{d}_{ij}$$

kašnjenje od ivice clock-a do pojave Q na izlazu FF_i :

minimalno δ_i , maksimalno: Δ_i

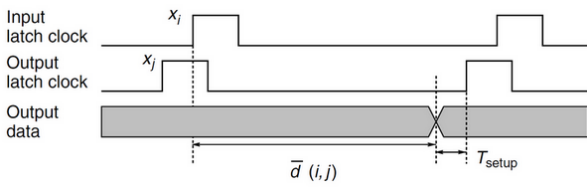
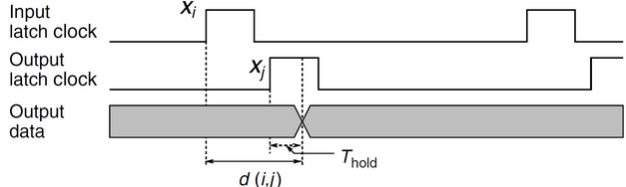
Setup time constraint	Hold time constraint
 $P \geq \Delta_i + \bar{d}_{ij} + T_{sj}$	 $\delta_i + \underline{d}_{ij} \geq T_{Hj}$

$$\underline{d}_{ij} \geq T_{Hj} - \delta_i$$

Ako je $\delta_i > T_{Hj}$ tada je Hold time constraint uve zadovoljen

c) Sekvencijalna kola - clock skew generalizacija za setup/hold

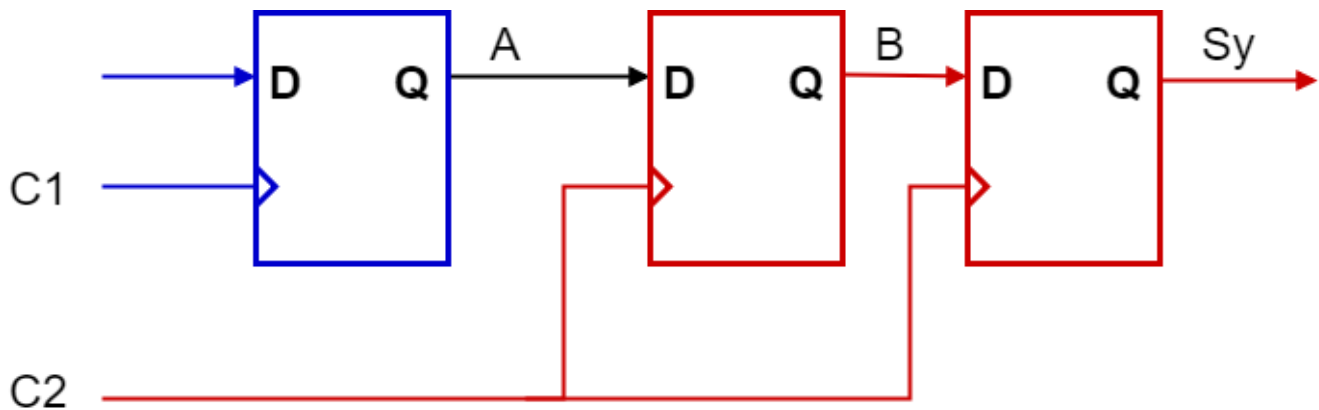
clock skew - kašnjenje rastuće ivice takta na FF-ovima

T setup	T hold
 $P \geq x_i - x_j + \bar{d}_{ij} + T_{setup}$	 $x_i + \underline{d}_{ij} \geq x_j + T_{hold}$

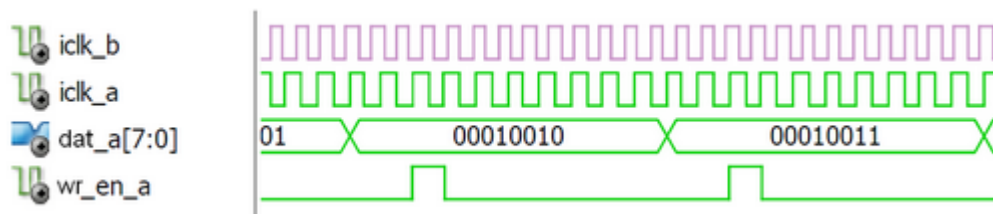
21. Prelazak između takt domena

a) Multi flop sinhronizator

Nestabilan signal (B) propuštamo kroz FF koji radi na istom clock-u.

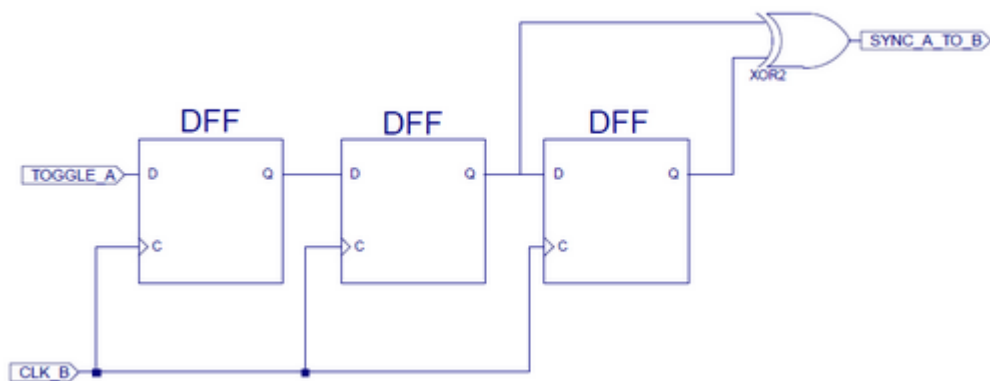


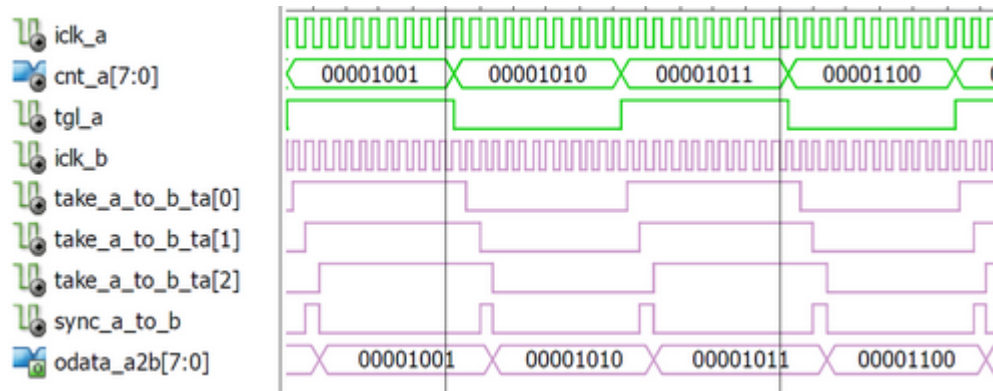
b) Transfer više bitnog vektora korišćenjem signala dozvole - četvrtka širine jedne periode takta



c) Transfer više bitnog vektora korišćenjem signala dozvole čija vrednost se menja za svaki novi podatak - toggle princip

Relativno spor tok podataka





```
sync_a_to_b <= take_a_to_b_ta(2) xor taek_a_to_b_ta(1)
```

d) Upotreba FIFO memorije kao bafera između različitih takt domena

Relativno brz tok podataka

